

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -2



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Recap of Previous Lecture



Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling

Topics to be Covered



Topic

SRTF Scheduling

Topic

LF Scheduling

Topic

LRTF Scheduling

Topic

HRRN Scheduling



Topic : SRTF (Shortest Remaining Time First)

preemptive version of SJF

Scheduling Criteria: smaller BT / Tie breaker $\Rightarrow$ FCFS

Type of Algorithm: preemptive

If a later arriving process smaller BT than current running process's remaining time then current process is preempted and new arrived process is scheduled.



Topic : SRTF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time	RT
P1	0	4	7	7	3	0
P2	1	2	3	2	0	0
P3	2	1	4	2	1	1

	BT
P1	4 3
P2	2 *
P3	1

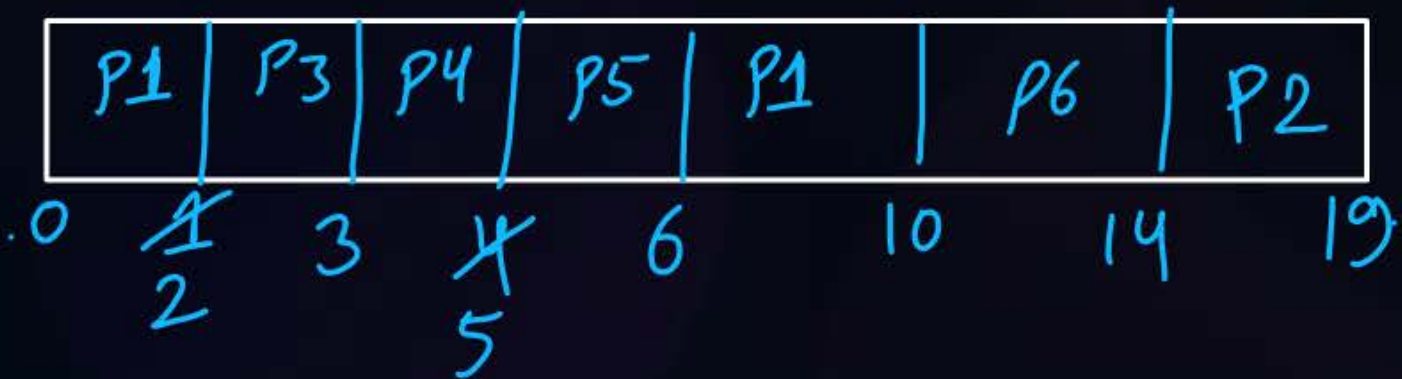
	Ready Queue
0	P1
1	P1, P2
2	P1, P2 , P3





Topic : SRTF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6	10		
P2	1	5	19		
P3	2	1	3		
P4	3	2	5		
P5	4	1	6		
P6	5	4	14		

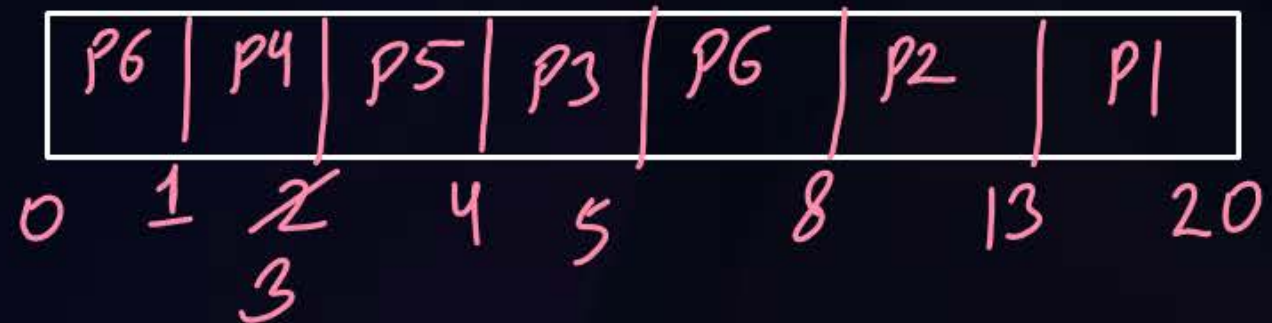


no. of context switches = 6



Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	4	7	20		
P2	5	5	13		
P3	3	1	5		
P4	1	2	3		
P5	2	1	4		
P6	0	4	8		





Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	9	22		
P2	1	6	14		
P3	2	4	9		
P4	3	2	5		
P5	6	1	7		

RT

0

0

0

0

0

P1	P2	P3	P4	P3	P5	P3	P2	P1	
0	1	2	3	5	6	7	9	14	22



Topic : SRTF (Shortest Remaining Time First)

Advantages:

and min avg TAT

1. Minimum average waiting time among all scheduling algorithm
2. Better throughput in continue run

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. Longer Processes may suffer from starvation

[NAT]

#Q. Response time of processes in non-preemptive scheduling algorithms are equal to waiting time of processes?

☒ True or False

Justify your answer with appropriate explanation.





Topic : LJF (Longest Job First)

longest [↓] burst time process first

Scheduling Criteria: Largest BT | Tie breaker $\Rightarrow$ FCFS

Type of Algorithm: Non-preemptive

- always suffers from convoy effect
- smaller processes may suffer from starvation



Topic : LJF (Longest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6	6		
P2	0	3	13		
P3	0	4	10		
P4	0	2	15		





Topic : LJF (Longest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	2	2		
P2	1	3	9		
P3	2	4	6		
P4	3	2	11		





Topic : LRTF (Longest Remaining Time First)

↓
preemptive version of LTF

Scheduling Criteria: Largest BT | Tie breaker $\Rightarrow$ FCFS

Type of Algorithm: Preemptive



Process	AT	BT
P1	0	2
P2	0	4
P3	0	3

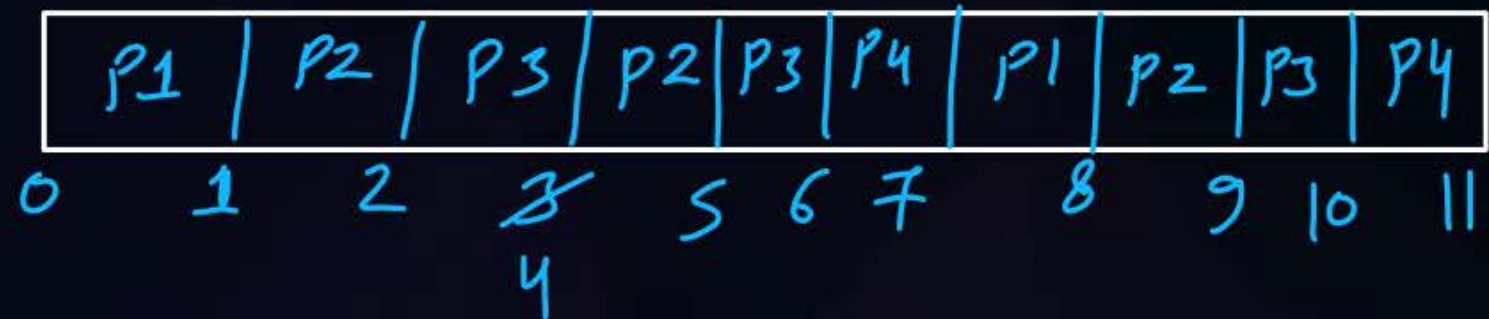
P1 ~~2~~
 P2 ~~4 3 2~~
 P3 ~~3 2~~

P2	P3	P1	P2	P3	P1	P2	P3	
0	1 2	3	4	5	6	7	8	9



Topic : LRTF (Longest Remaining Time First)

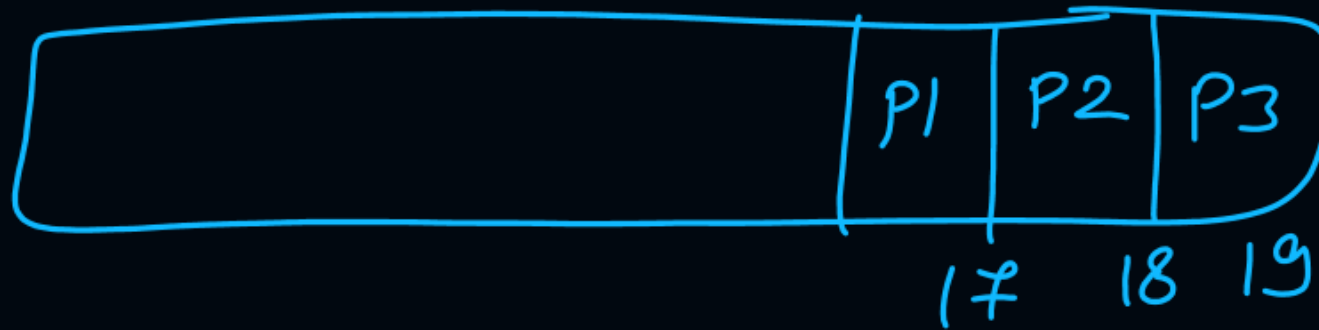
Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	2			
P2	1	3			
P3	2	4			
P4	3	2			



Ques)

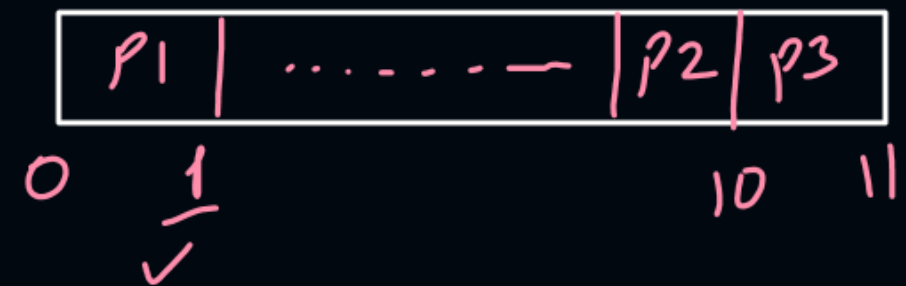
P1	0	9
P2	0	6
P3	0	4

if no any process completes
before all processes arrive



Ques)

	AT	BT	CT
P1	0	1	1
P2	1	6	10
P3	2	4	11



[MCQ]

GATE-PYQ



#Q. Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is:

A 13 units

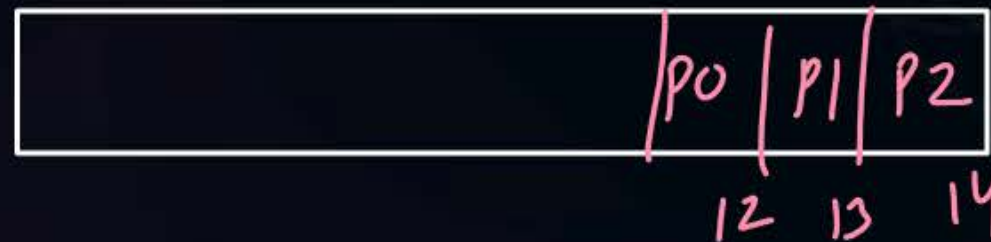
B 14 units

C 15 units

D 16 units

	AT	BT	CT	TAT
P0	0	2	12	12
P1	0	4	13	13
P2	0	8	14	14

$$\text{avg TAT} = \frac{12+13+14}{3} = 13$$



H.W.

Ques 1)

	AT	BT
P1	0	6
P2	1	2
P3	3	9
P4	2	5

FCFS, SJF, SRTF

Ques 2)

	AT	BT
P1	0	8
P2	2	4
P3	8	2
P4	1	6

SRTF, SJF, FCFS

Vishvadeepsir



2 mins Summary

Topic

SRTF Scheduling

Topic

LJF Scheduling

Topic

LRTF Scheduling

Topic

HRRN Scheduling



Happy Learning

THANK - YOU

